Practical Work Report (TP)

Evaluation and Comparison of Search Algorithms

1. Introduction & Objectives

The objective of this practical work is to evaluate and compare the performance of four fundamental search algorithms: Simple Sequential Search, Optimised Sequential Search (for sorted tables), Iterative Binary Search, and Recursive Binary Search. The comparison relies on two core criteria: (a) Number of Comparisons and (b) Execution Time.

2. Experimental Methodology

Following the lab guidelines, large arrays of random elements were generated with sizes of N = 10^3, 10■, and 10■. To ensure statistical accuracy and eliminate anomalies, 30 independent tests were conducted for each case, and the average values were recorded using high-precision timers.

3. Summary of Findings & Analysis

• **Sequential Search:** Exhibits linear time complexity O(N), making performance degrade rapidly for large arrays (10■).
• **Optimised Sequential Search:** Improves performance on sorted arrays by terminating early, but remains bounded by O(N).
• **Binary Search (Iterative & Recursive):** Demonstrates logarithmic time complexity O(log N), maintaining extreme speed and minimal comparison counts even for massive datasets.

4. Conclusion

The experiments confirm that algorithm selection is critical for application performance. While sequential searches are sufficient for small datasets, binary search is vastly superior and essential for large-scale searching operations.

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